First-order fluctuations of small-γ Liouville quantum gravity distance

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Abstract

Let h be a whole-plane Gaussian free field with $h_1(0) = 0$. For each $\gamma \in (0, 2)$, let D_h^γ denote the γ-LQG metric of [GM21, Theorem 1.1], with the metrics for different values of γ defined using the same field. We prove that

$$\frac{D_h^\gamma(0,1) - 1}{\gamma} \xrightarrow{\mathbb{P}} \frac{1}{2}\left(\int_0^1 h(x)\,dx - q\right), \qquad q = \operatorname{med}\left(\min_{y\in[0,1]} \int_0^1 h(x+iy)\,dx\right).$$

The integrals are defined as L^2 limits of mollifications. The limiting law is $\mathcal{N}(-q/2, 3/8)$ and $q < 0$. The proof uses moment estimates which are uniform for small γ, differentiation in Cameron–Martin directions, and convergence in logarithmic energy of the D_h^γ-length measures of geodesics. The parameter dependence in the LFPP estimates used to obtain the uniform metric bounds is recorded in Appendix A.

1 Statement and notation

We identify $\mathbb{C}$ with $\mathbb{R}^2$ and write $B_r(z)$ for the Euclidean ball of radius r centered at z. The covariance kernel of h is

$$G(z, w) = -\log|z - w| + \log|z|_+ + \log|w|_+, \qquad |z|_+ = \max\{1, |z|\}. \tag{1.1}$$

This is an identity for a generalized function, not a definition of pointwise values of h. The circle average of h on $\partial B_r(z)$ is denoted by $h_r(z)$.

For each $\gamma \in (0, 2)$, let D_h^γ denote the γ-LQG metric of [GM21], viewed as a measurable function of h. We use the normalization of the limiting metric in [GM21, Theorem 1.1 and the proof of Theorems 1.1, 1.2, and 1.8]: for a whole-plane GFF with $h_1(0) = 0$, the median D_h^γ-distance between the left and right boundaries of the unit square is one. Set

$$\xi_\gamma = \frac{\gamma}{d_\gamma}, \qquad Q_\gamma = \frac{2}{\gamma} + \frac{\gamma}{2}. \tag{1.2}$$

The dimension bounds of [DG20, Theorem 1.2] imply

$$d_\gamma = 2 + O(\gamma), \qquad \frac{\xi_\gamma}{\gamma} \longrightarrow \frac{1}{2}, \qquad \xi_\gamma Q_\gamma = 1 + O(\xi_\gamma). \tag{1.3}$$

We use existence, uniqueness, Weyl scaling, affine covariance, and LFPP convergence from [DFG$^+$20, GM21]. The estimates which are uniform as $\gamma \to 0$ are proved in Appendix A.

For sets $A, B \subset \mathbb{C}$, write

$$D_h^\gamma(A, B) = \inf_{z\in A, w\in B} D_h^\gamma(z, w).$$

The notation $D_h^\gamma(z,w;U)$ denotes the internal metric, with paths restricted to U. These are different quantities. We write

$$C_\gamma := \inf_{s,t\in[0,1]} D_h^\gamma(is, 1+it). \tag{1.4}$$

A median is the lower 1/2-quantile. No path restriction is imposed in (1.4).

For distinct $z, w \in \mathbb{C}$, define

$$\lambda_{z,w} := \int_0^1 \delta_{(1-u)z+uw}\, du. \tag{1.5}$$

The pairing $(h, \lambda_{z,w})$ is the L^2 limit under smooth mollification. For the crossing endpoints, write

$$H(s,t) := (h, \lambda_{is,1+it}), \qquad s,t \in [0,1], \qquad M := \min_{y\in[0,1]} H(y,y), \qquad q := \operatorname{med} M. \tag{1.6}$$

We prove below that H has a continuous version and that M has a strictly positive continuous density.

Theorem 1.1. *As $\gamma \to 0$,*

$$\frac{D_h^\gamma(0,1) - 1}{\gamma} \xrightarrow{\mathbb{P}} \frac{H(0,0) - q}{2}. \tag{1.7}$$

Here $H(0,0) = \int_0^1 h(x)\, dx$. Consequently,

$$\frac{D_h^\gamma(0,1) - 1}{\gamma} \xrightarrow{\text{law}} \mathcal{N}\left(-\frac{q}{2}, \frac{3}{8}\right). \tag{1.8}$$

Moreover, $q < 0$. If $m_\gamma = \operatorname{med} D_h^\gamma(0,1)$, then

$$m_\gamma = 1 - \frac{q}{2}\gamma + o(\gamma). \tag{1.9}$$

Constants and moment orders. All moment orders are fixed before $\gamma \to 0$. A sufficiently small upper bound for γ may depend on the moment order. Constants may depend on a fixed compact set and on a fixed positive lower bound for endpoint separation, but not on γ in the indicated range. We do not assert one positive upper bound for γ which works simultaneously for all moment orders.

2 Moment estimates and length measures of geodesics

For a compact set $K \subset \mathbb{C}$ and $\delta > 0$, let

$$A_\gamma(K) := \sup_{\substack{z,w\in K \\ z\neq w}} \frac{D_h^\gamma(z,w)}{|z-w|^{1/2}}, \tag{2.1}$$

$$I_\gamma(K,\delta) := \inf_{\substack{z,w\in K \\ |z-w|\geq\delta}} D_h^\gamma(z,w). \tag{2.2}$$

An empty supremum in (2.1) is zero. If the index set in (2.2) is empty, its inverse is defined to be zero.

Lemma 2.1. *For every $p \in [1,\infty)$ there exists $\gamma_p > 0$ such that, for each compact K and $\delta > 0$,*

$$\sup_{0<\gamma\leq\gamma_p} \mathbb{E}\left[A_\gamma(K)^p + I_\gamma(K,\delta)^{-p}\right] < \infty. \tag{2.3}$$

For $0 < \delta \leq 1$, the second expectation is at most $C_{K,p}\delta^{-c_p}$, with constants independent of γ.

Proof. See Appendix A. □

Let $\mathcal{H}$ be the Cameron–Martin space of the pinned whole-plane GFF. Its elements have zero unit-circle average and inner product

$$(f,g)_\nabla = \frac{1}{2\pi}\int_{\mathbb{C}} \nabla f(z)\cdot\nabla g(z)\,d^2z.$$

For a compactly supported finite signed measure μ whose total variation has finite logarithmic energy, its squared dual norm is

$$\|\mu\|^2_{\mathcal{H}^*} = \iint G(z,w)\,\mu(\,dz)\mu(\,dw). \tag{2.4}$$

For a measure of nonzero total mass this means the functional on pinned representatives. For a random measure the identity is applied to each realization, not as a conditional covariance identity. Equivalently, one can replace μ by $\mu-\mu(\mathbb{C})\sigma$, where σ is uniform probability measure on $\partial B_1(0)$. The two measures have the same pairing with each element of $\mathcal{H}$.

We also use that geodesics exist. For small γ, this follows directly from Lemma 2.1. Indeed, the scaling calculation (2.9) below and Borel–Cantelli give

$$D_h^\gamma(\partial B_{2^k}(0),\partial B_{2^{k+1}}(0)) \longrightarrow \infty \qquad \text{a.s.}$$

A path to a sufficiently distant point must cross one of these annuli. Hence every closed bounded D_h^γ-ball is contained in a Euclidean compact set. Since D_h^γ induces the Euclidean topology, $(\mathbb{C}, D_h^\gamma)$ is boundedly compact. Since it is a length space, it is geodesic. Constant-speed geodesics between fixed endpoints form a compact subset of $C([0,1],\mathbb{C})$.

If P is a D_h^γ-geodesic from z to w, parameterized at constant metric speed on $[0,1]$, its D_h^γ-length probability measure is

$$\nu_P := \int_0^1 \delta_{P(t)}\,dt. \tag{2.5}$$

For fixed deterministic endpoints, the set of such geodesics is nonempty and compact. Its graph, as a function of the continuous metric, is Borel. We can therefore choose one such geodesic measurably; write $\nu^\gamma_{z,w}$ for its length measure. Every estimate below is uniform over the choice of geodesic, so the particular measurable choice will not matter.

Lemma 2.2. *Fix a compact set K and $\delta>0$. There are measurable random variables $T_\gamma\ge 2$ and $B_\gamma\ge 1$ such that every geodesic between $z,w\in K$ with $|z-w|\ge\delta$ satisfies*

$$\operatorname{supp}\nu_P \subset B_{T_\gamma}(0), \qquad \nu_P(B_r(u))\le B_\gamma r^{1/2}, \qquad u\in\mathbb{C},\quad 0<r\le 1. \tag{2.6}$$

For each fixed $p\in[1,\infty)$, the pth moments of T_γ, B_γ, and a common upper bound for $\|\nu_P\|_{\mathcal{H}^*}$ are bounded uniformly for sufficiently small γ.

Proof. Write $h^R = h(R\,\cdot) - h_R(0)$. Affine covariance and Weyl scaling give

$$D_h^\gamma(Rx,Ry) = R^{\xi_\gamma Q_\gamma} e^{\xi_\gamma h_R(0)} D_{h^R}^\gamma(x,y). \tag{2.7}$$

The field h^R has the law of h. It need not be independent of $h_R(0)$. Since $\operatorname{Var} h_R(0) = \log R$ for $R\ge 1$, Cauchy–Schwarz and Lemma 2.1 give

$$\mathbb{E}[A_\gamma(B_R(0))^p] \le C_p R^{p(\xi_\gamma Q_\gamma - 1/2)+p^2\xi_\gamma^2/2}, \tag{2.8}$$

$$\mathbb{E}[L_R^{-p}] \le C_p R^{-p\xi_\gamma Q_\gamma + p^2\xi_\gamma^2/2}, \qquad L_R := D_h^\gamma(\partial B_R(0), \partial B_{2R}(0)). \tag{2.9}$$

The lower bound uses global set distance, so it follows from (2.3) applied to the two rescaled circles.

Choose a deterministic dyadic $R_0 \ge 2$ with $K \subset B_{R_0}(0)$ and put $U_\gamma = \operatorname{diam}(K; D_h^\gamma)$. If a geodesic with endpoints in K exits $B_{2R}(0)$, where $R \ge R_0$, its length is at least L_R and at most U_γ. For each fixed N, choose a sufficiently high fixed moment order in (2.9) and in the moment bound for U_γ. Since $\xi_\gamma Q_\gamma \to 1$,

$$\mathbb{P}[L_R \le U_\gamma] \le \mathbb{P}[U_\gamma > R^{1/4}] + \mathbb{P}[L_R \le R^{1/4}] \le C_N R^{-N} \tag{2.10}$$

for sufficiently small γ. Define T_γ to be twice the first dyadic $R \ge R_0$ for which $L_R > U_\gamma$. Then T_γ is finite a.s., and (2.10) gives all of its required moments. Every geodesic under consideration is contained in $B_{T_\gamma}(0)$.

Consider the first and last visits of a geodesic to a closed Euclidean ball of radius r. The subpath between them is a geodesic and its length is bounded by $A_\gamma(B_{T_\gamma+1}(0))(2r)^{1/2}$. Consequently,

$$\nu_P(B_r(u)) \le \frac{\sqrt{2}\, A_\gamma(B_{T_\gamma+1}(0))}{I_\gamma(K,\delta)} r^{1/2}. \tag{2.11}$$

Repeated visits to the ball cause no difficulty. Decompose according to the dyadic size of T_γ, use (2.8), and then use (2.10) at an arbitrarily high fixed order. Cauchy–Schwarz and the inverse moments of $I_\gamma(K,\delta)$ show that the coefficient in (2.11), enlarged to be at least one, has every required fixed moment.

For any probability measure satisfying (2.6),

$$\begin{aligned}\iint \log^+ \frac{1}{|z-w|}\, \nu_P(\,dz)\nu_P(\,dw) &= \int_0^\infty (\nu_P \otimes \nu_P)\{|z-w| < e^{-t}\}\, dt \\ &\le \int_0^\infty \min\{1, B_\gamma e^{-t/2}\}\, dt \le 2\log B_\gamma + 2.\end{aligned}$$

Together with (1.1), this gives

$$\|\nu_P\|_{\mathcal{H}^*}^2 \le 2\log B_\gamma + 2 + 2\log T_\gamma. \tag{2.12}$$

□

3 Differentiation in Cameron–Martin directions

Choose an orthonormal basis $(e_j)_{j\ge1}$ of $\mathcal{H}$ consisting of smooth bounded functions. Such a basis is obtained from a countable dense family of compactly supported smooth functions by subtracting unit-circle averages and applying Gram–Schmidt. Write X_j for the associated independent standard Gaussian coordinates of h.

Lemma 3.1. *Fix deterministic $z \ne w$ and γ sufficiently small for the moment estimates used below. Along almost every Gaussian coordinate line, $\log D_h^\gamma(z,w)$ has a Lipschitz representative. Its coordinate derivatives satisfy, a.s. and simultaneously for all j,*

$$\partial_j \log D_h^\gamma(z,w) = \xi_\gamma \int e_j\, d\nu_P \tag{3.1}$$

for every D_h^γ-geodesic P from z to w.

Proof. Weyl scaling gives

$$\left|\log D^{\gamma}_{h+te_j}(z,w) - \log D^{\gamma}_{h+se_j}(z,w)\right| \le \xi_\gamma \|e_j\|_\infty |t-s|.$$

Apply Fubini first to the field and the shift and then to the remaining Gaussian coordinates. This gives a Lipschitz representative on almost every coordinate line, hence differentiability at almost every point of almost every line. Since the basis is countable, another application of Fubini gives a full-probability set on which the coordinate derivatives exist simultaneously for every j.

At such a field, fix any D^{γ}_h-geodesic P. It is a competitor for the Weyl-scaled metric. Hence, for almost every sufficiently small t,

$$\log D^{\gamma}_{h+te_j}(z,w) - \log D^{\gamma}_h(z,w) \le \log \int e^{\xi_\gamma t e_j}\, d\nu_P = \xi_\gamma t \int e_j\, d\nu_P + O(t^2).$$

Divide first by positive and then by negative t, and take limits through shifts for which the Weyl identity holds. The resulting upper and lower bounds agree. This proves (3.1). The argument uses only that P is a geodesic for the unshifted metric, so it applies to every such P. □

We use the following Gaussian L^p inequality. If F is coordinatewise Lipschitz, $F \in L^p$, and its weak coordinate derivatives satisfy the indicated integrability, then

$$\|F - \mathbb{E}F\|_p \le c_p \left\| \left(\sum_{j\ge 1} |\partial_j F|^2 \right)^{1/2} \right\|_p, \qquad 1 < p < \infty. \tag{3.2}$$

For completeness, first suppose that F is smooth and depends on finitely many standard Gaussian variables. For independent Gaussian vectors X, Y, differentiate $F(X\cos t + Y \sin t)$ for $0 \le t \le \pi/2$. Its derivative is the inner product of the gradient with an independent standard Gaussian vector. Minkowski's inequality and conditional Jensen give (3.2), with

$$c_p = \frac{\pi}{2} \left(\mathbb{E}|Z|^p\right)^{1/p}, \qquad Z \sim \mathcal{N}(0,1).$$

For a general coordinatewise Lipschitz $F \in L^p$, set $F_n = \mathbb{E}[F \mid X_1, \dots, X_n]$. Gaussian integration by parts on each coordinate line gives $\partial_j F_n = \mathbb{E}[\partial_j F \mid X_1, \dots, X_n]$ for $j \le n$. Conditional Jensen controls the gradient norm of F_n by the conditional expectation of the full gradient norm. Finite-dimensional approximation followed by L^p martingale convergence proves (3.2). Bounds at $p = 1$ follow by using a larger moment order.

For a finite-energy measure μ,

$$\sum_{j\ge 1} \left(\int e_j\, d\mu \right)^2 = \|\mu\|^2_{\mathcal{H}^*}.$$

Therefore Lemmas 2.2–3.1 and (3.2) imply, with

$$b_\gamma := \mathbb{E} \log D^{\gamma}_h(0,1), \tag{3.3}$$

that

$$\|\log D^{\gamma}_h(0,1) - b_\gamma\|_p \le C_p \xi_\gamma. \tag{3.4}$$

Lemma 2.1 bounds $|b_\gamma|$ uniformly and supplies the logarithmic integrability used here.

4 Euclidean approximation and convergence in energy

For deterministic $z \neq w$, set $r = |w - z|$ and $\theta = \arg(w - z)$. The field

$$\widetilde{h}(u) = h(z + re^{i\theta}u) - h_r(z)$$

has the same law as h. Affine covariance gives

$$\log D_h^\gamma(z, w) = \xi_\gamma Q_\gamma \log r + \xi_\gamma h_r(z) + \log D_{\widetilde{h}}^\gamma(0, 1). \tag{4.1}$$

In particular,

$$\mathbb{E} \log D_h^\gamma(z, w) = b_\gamma + \xi_\gamma Q_\gamma \log |w - z|. \tag{4.2}$$

No independence is used in either identity.

Lemma 4.1. *For each $R < \infty$ and $p \in [1, \infty)$,*

$$\|E_\gamma(R)\|_p \le C_{R,p}\xi_\gamma^{1/2}, \qquad E_\gamma(R) := \sup_{z,w \in B_R(0)} \left| e^{-b_\gamma} D_h^\gamma(z, w) - |z - w| \right|. \tag{4.3}$$

Proof. For $r > 0$, put

$$X = \log D_h^\gamma(z, w) - b_\gamma - \xi_\gamma Q_\gamma \log r.$$

Equation (4.1), (3.4), and the circle-average covariance give

$$\|X\|_p \le C_{R,p}\xi_\gamma \sqrt{1 + \log^+(1/r)}.$$

Hölder's inequality, the Gaussian exponential moment for $h_r(z)$, and positive moments of $D_{\widetilde{h}}^\gamma(0, 1)$ give

$$\|e^X\|_{2p} \le C_{R,p} r^{-C_p \xi_\gamma^2}, \qquad 0 < r \le 1,$$

and a constant bound for $1 \le r \le 2R$. Since $|e^x - 1| \le |x|(1 + e^x)$,

$$\left\| e^{-b_\gamma} D_h^\gamma(z, w) - r^{\xi_\gamma Q_\gamma} \right\|_p \le C_{R,p}\xi_\gamma r^{\xi_\gamma Q_\gamma - C_p \xi_\gamma^2} \sqrt{1 + \log^+(1/r)} \le C_{R,p}\xi_\gamma. \tag{4.4}$$

The last bound follows from $\xi_\gamma Q_\gamma \to 1$. Also,

$$\sup_{0 \le r \le 2R} |r^{\xi_\gamma Q_\gamma} - r| \le C_R \xi_\gamma$$

by (1.3).

A grid of mesh ξ_γ in a slightly larger ball has $O_R(\xi_\gamma^{-4})$ ordered endpoint pairs. Apply (4.4) at one fixed moment order $P \ge \max\{p, 8\}$. The maximum grid error has L^p norm at most $C\xi_\gamma^{1-4/P}$. The triangle inequality and Lemma 2.1, together with the boundedness of b_γ, fill the grid with error at most $C\xi_\gamma^{1/2}$ in L^p. □

Lemma 4.2. *For deterministic endpoints in a fixed compact set with $|z - w| \ge \delta > 0$,*

$$\left\| \, \|\nu_{z,w}^\gamma - \lambda_{z,w}\|_{\mathcal{H}^*} \, \right\|_p \le C_{K,\delta,p}\xi_\gamma^{1/24}. \tag{4.5}$$

The bound is uniform over deterministic endpoint pairs and measurable choices of geodesic.

Proof. Choose a fixed ball much larger than K. If $E_\gamma(R)$ is smaller than a sufficiently small fixed constant, every geodesic with endpoints in K stays inside this ball. Otherwise, the first hit of the boundary would have larger $e^{-b_\gamma}D_h^\gamma$-distance from the initial endpoint than the whole geodesic. By Lemma 4.1, the complement of this event has probability at most $C_N\xi_\gamma^N$ for every fixed N, after decreasing the upper bound for γ.

Parameterize a geodesic P at constant metric speed on $[0,1]$. Then

$$e^{-b_\gamma}D_h^\gamma(z,P(t)) = te^{-b_\gamma}D_h^\gamma(z,w), \qquad e^{-b_\gamma}D_h^\gamma(P(t),w) = (1-t)e^{-b_\gamma}D_h^\gamma(z,w).$$

The identity

$$|v-((1-t)z+tw)|^2 = (1-t)|v-z|^2 + t|v-w|^2 - t(1-t)|z-w|^2$$

and the uniform Euclidean approximation give

$$\sup_{t\in[0,1]} |P(t)-((1-t)z+tw)| \le CE_\gamma(R)^{1/2} =: \eta. \tag{4.6}$$

Thus $\nu := \nu_{z,w}^\gamma$ and $\lambda := \lambda_{z,w}$ admit a coupling with displacement at most η.

For $0<\rho<1$, replace $-\log|u-v|$ by $-\log\max\{|u-v|,\rho\}$. This kernel is $1/\rho$-Lipschitz in either variable. The one-variable terms in (1.1) cancel since $(\nu-\lambda)(\mathbb{C})=0$. The truncated bilinear form is bounded in absolute value by $C\eta/\rho$. The self-energy truncation error for ν is at most

$$\int_0^\rho B_\gamma r^{1/2}\frac{dr}{r} = 2B_\gamma\rho^{1/2}.$$

The measure λ satisfies $\lambda(B_r(u)) \le 2r/\delta$, so it has the same type of bound with a deterministic constant. The mixed truncation error has negative coefficient and can be discarded in an upper bound. Hence

$$\|\nu-\lambda\|_{\mathcal{H}^*}^2 \le C\eta/\rho + C(B_\gamma+1)\rho^{1/2}. \tag{4.7}$$

Shrink the fixed threshold of the good event so that $\eta<1$ and take $\rho=\eta^{2/3}$. The case $\eta=0$ follows by a limit. We obtain

$$\|\nu-\lambda\|_{\mathcal{H}^*} \le C(B_\gamma+1)^{1/2}E_\gamma(R)^{1/12}.$$

Hölder's inequality and Lemmas 2.2 and 4.1 give (4.5) on this event. On the complement, use uniform higher moments of $\|\nu\|_{\mathcal{H}^*}+\|\lambda\|_{\mathcal{H}^*}$ and the bound $C_N\xi_\gamma^N$ with N sufficiently large. □

5 First-order expansion and the crossing distance

Proposition 5.1. *For $p\in[1,\infty)$ and deterministic endpoints in a fixed compact set with $|z-w|\ge\delta>0$,*

$$\left\| \frac{\log D_h^\gamma(z,w) - b_\gamma - \xi_\gamma Q_\gamma \log|z-w|}{\xi_\gamma} - (h,\lambda_{z,w}) \right\|_p \le C_{K,\delta,p}\xi_\gamma^{1/24}. \tag{5.1}$$

Proof. The expression has mean zero by (4.2). Its jth Gaussian coordinate derivative is

$$\int e_j\, d(\nu_{z,w}^\gamma - \lambda_{z,w})$$

by Lemma 3.1. Apply (3.2) and Lemma 4.2. At $p=1$, use a larger moment order. □

Lemma 5.2. *The centered Gaussian process $H(s,t)$ of (1.6) has a continuous version on $[0,1]^2$.*

Proof. The segments have Euclidean length at least one, so their probability measures satisfy $\lambda(B_r(u)) \le 2r$. Put

$$a = |s - s'| \vee |t - t'| \le 1.$$

Changing the two endpoint parameters by at most a gives a coupling of displacement at most Ca. The truncation argument in (4.7), now with linear ball-mass bounds, gives

$$\mathbb{E}[|H(s,t) - H(s',t')|^2] \le Ca^{1/2}$$

by taking $\rho = a^{1/2}$. Gaussian higher moments and the two-parameter Kolmogorov continuity criterion give any Hölder exponent strictly less than $1/4$. The same finite-energy estimates justify the L^2 definition in (1.5). □

Proposition 5.3. *Let $r(s,t) = \sqrt{1 + (s-t)^2}$. Then*

$$\sup_{s,t\in[0,1]} \left| \frac{\log D_h^\gamma(is, 1+it) - b_\gamma - \xi_\gamma Q_\gamma \log r(s,t)}{\xi_\gamma} - H(s,t) \right| \xrightarrow{\mathbb{P}} 0. \tag{5.2}$$

Proof. Take a grid of mesh ξ_γ^3 in the two endpoint parameters, including the boundary. It has $O(\xi_\gamma^{-6})$ points. Proposition 5.1, Markov's inequality, and a union bound show that the probability of a residual exceeding ε at a grid point is at most

$$C_p \varepsilon^{-p} \xi_\gamma^{p/24-6}.$$

Fix one $p > 144$ and then let $\gamma \to 0$.

For interpolation, the triangle inequality gives

$$|\log D(z,w) - \log D(z',w')| \le \frac{D(z,z') + D(w,w')}{\min\{D(z,w), D(z',w')\}}.$$

The endpoints remain on opposite sides of the square. Lemma 2.1 makes the interpolation error, after division by ξ_γ, at most a tight random factor times $\xi_\gamma^{1/2}$. The deterministic term contributes $O(\xi_\gamma^2)$ since $\xi_\gamma Q_\gamma$ is bounded. The interpolation error for H tends to zero a.s. by Lemma 5.2. □

Taking the infimum in (5.2) gives

$$\frac{\log C_\gamma - b_\gamma}{\xi_\gamma} = \inf_{s,t\in[0,1]} \left\{ Q_\gamma \log \sqrt{1 + (s-t)^2} + H(s,t) \right\} + o_{\mathbb{P}}(1). \tag{5.3}$$

The deterministic term is nonnegative, vanishes on $s = t$, and tends uniformly to infinity outside every fixed neighborhood of the diagonal. For every continuous realization of H, compactness therefore gives convergence of the infimum to M. Hence

$$\frac{\log C_\gamma - b_\gamma}{\xi_\gamma} \xrightarrow{\mathbb{P}} M. \tag{5.4}$$

At the fixed endpoint pair $(0,1)$, Proposition 5.1 gives

$$\frac{\log D_h^\gamma(0,1) - b_\gamma}{\xi_\gamma} \xrightarrow{\mathbb{P}} H(0,0). \tag{5.5}$$

6 Medians and the limiting Gaussian

Lemma 6.1. *The law of M has a strictly positive continuous density on $\mathbb{R}$. It has a unique median q, and $q < 0$.*

Proof. Choose a smooth compactly supported function f which is one on a neighborhood of $[0,1]^2$ and has zero unit-circle average. This can be done by subtracting a suitable smooth bump supported near an arc of the unit circle disjoint from the square. Thus $f \in \mathcal{H}$ and $\|f\|_{\mathcal{H}} > 0$. Write

$$h = X\frac{f}{\|f\|_{\mathcal{H}}} + h^{\perp},$$

where $X \sim \mathcal{N}(0,1)$ is independent of $h^{\perp}$. The line integral of f along every horizontal unit segment in the square is one, so

$$M = \frac{X}{\|f\|_{\mathcal{H}}} + \min_{y\in[0,1]} H^{\perp}(y,y).$$

The second term is finite a.s. Convolution with a nondegenerate Gaussian gives an everywhere positive continuous density.

The pair $(H(0,0), H(1,1))$ is a nondegenerate centered Gaussian pair. Indeed, smooth functions supported near interior points of the two segments and away from the unit circle can vary the two line integrals independently. Therefore

$$\mathbb{P}[M \geq 0] \leq \mathbb{P}[H(0,0) \geq 0, H(1,1) \geq 0] < \frac{1}{2}.$$

Since the distribution function of M is strictly increasing, its median satisfies $q < 0$. □

Proof of Theorem 1.1. By our normalization of D_h^{γ}, $\operatorname{med} C_{\gamma} = 1$. Hence zero is a median of $\log C_{\gamma}$, so $-b_{\gamma}/\xi_{\gamma}$ is a median of the left side of (5.4). Weak convergence to a random variable with continuous strictly increasing distribution function implies convergence of these medians. Lemma 6.1 gives

$$\frac{b_{\gamma}}{\xi_{\gamma}} \longrightarrow -q. \tag{6.1}$$

Combining this with (5.5),

$$\frac{\log D_h^{\gamma}(0,1)}{\xi_{\gamma}} \xrightarrow{\mathbb{P}} H(0,0) - q. \tag{6.2}$$

In particular, $\log D_h^{\gamma}(0,1) = O_{\mathbb{P}}(\xi_{\gamma})$. The expansion $e^x - 1 = x + O(x^2 e^{|x|})$ and $\xi_{\gamma}/\gamma \to 1/2$ prove (1.7).

For $u, v \in [0,1]$, (1.1) reduces to $-\log|u-v|$. Consequently,

$$\operatorname{Var} H(0,0) = \int_0^1\int_0^1 -\log|u-v|\,du\,dv = 2\int_0^1 (1-r)(-\log r)\,dr = \frac{3}{2}. \tag{6.3}$$

This proves (1.8). Median convergence in (5.5) gives $(\log m_{\gamma} - b_{\gamma})/\xi_{\gamma} \to 0$, since the limiting Gaussian has unique median zero. Together with (6.1), this proves (1.9). □

A Uniform moment estimates for small γ

We prove Lemma 2.1. We start from the heat-kernel white-noise approximation and use the geometric estimates of [DDDF20]. The estimates in [DDDF20] are stated with ξ fixed. What is needed here is that the constants can be chosen uniformly when ξ ranges over a sufficiently small interval $(0, \xi_*]$. We therefore record the parameter dependence in the parts of the argument which are used below. Equation and proposition numbers from [DDDF20] refer to arXiv version 2.

Write

$$p_t(z) = \frac{1}{2\pi t} \exp\left(-\frac{|z|^2}{2t}\right), \qquad \phi_{0,n}(z) = \sqrt{\pi} \int_{2^{-2n}}^{1} \int_{\mathbb{R}^2} p_{t/2}(z-w)\, W(\,dw,\, dt).$$

The finite-range approximation from [DDDF20, Section 2] is denoted by $\psi_{0,n}$. Fix its cutoff function and range parameters once and for all. For a rectangle R with two marked opposite sides, let $L_n(R;\phi)$ be its internal crossing length for $e^{\xi\phi_{0,n}}|\,dz|$. Define $L_n(R;\psi)$ similarly. Here ξ is initially an independent parameter in $(0, \xi_*]$; later it is set equal to ξ_γ. Let λ_n be the median internal crossing length of the unit square for $\phi_{0,n}$, with $\lambda_0 = 1$.

A.1 Gaussian and quantile estimates

For a fixed rectangle V, Gaussian maximum estimates and Gaussian concentration give

$$\mathbb{E} \sup_V |\phi_{0,k}| \le C_V(k+1), \qquad \left\| e^{\xi \sup_V |\phi_{0,k}|} \right\|_p \le \exp\{C_V \xi (k+1) + C_V p \xi^2 (k+1)\}. \tag{A.1}$$

On fixed-size translates of V, the constants are the same since ϕ is stationary. The scaled gradient maximum $2^{-k} \sup_V |\nabla \phi_{0,k}|$ has mean at most $C_V\sqrt{k+1}$ and has a Gaussian upper tail about its mean with constants independent of k.

Proposition 5 of [DDDF20] constructs a coupling for which

$$\mathbb{P}\left[\sup_{n\ge 0} \|\phi_{0,n} - \psi_{0,n}\|_{L^\infty(V)} > t\right] \le C_V e^{-c_V t^2}. \tag{A.2}$$

The estimates in (A.1)–(A.2) concern the Gaussian fields before multiplication by ξ, so their constants are independent of ξ.

We use the following elementary quantile comparison. If positive random variables U, V satisfy

$$e^{-\xi Z} V \le U \le e^{\xi Z} V, \qquad \mathbb{P}[Z > t] \le \varepsilon,$$

then, whenever $0 < \varepsilon < p < 1 - \varepsilon$,

$$e^{-\xi t} q_{p-\varepsilon}(V) \le q_p(U) \le e^{\xi t} q_{p+\varepsilon}(V). \tag{A.3}$$

Here q_p is the lower p-quantile. This follows directly from the two corresponding event inclusions and does not use independence.

Set

$$\ell_n(\phi, p) = q_p(L_n([0,1]^2;\phi)), \qquad \overline{\ell}_n(\phi, p) = q_{1-p}(L_n([0,1]^2;\phi)),$$

and

$$\Lambda_n(\phi, p) = \max_{0\le j\le n} \frac{\overline{\ell}_j(\phi, p)}{\ell_j(\phi, p)}.$$

Then, for each fixed $p \in (0, 1/2)$,

$$\Lambda_n(\phi, p) \le C_p \Lambda_n(\psi, p/2), \qquad \Lambda_n(\psi, p) \le C_p \Lambda_n(\phi, p/2), \tag{A.4}$$

uniformly for $\xi \le \xi_*$. We will use this comparison only for finitely many fixed quantile probabilities.

A.2 Uniform crossing tails

The next lemma is the uniform form of the crossing-tail input from [DDDF20] which will be used in the rest of the appendix.

Lemma A.1. *There are $\xi_* > 0$, $p_0 \in (0, 1/2)$, and constants $c, C, u_0 > 0$ such that, for every $\xi \in (0, \xi_*]$, every $n \geq 0$, and every rectangle R in a fixed finite collection of shapes and marked directions,*

$$\mathbb{P}[L_n(R;\phi) < e^{-u}\ell_n(\phi, p_0)] \leq Ce^{-cu^2}, \tag{A.5}$$

$$\mathbb{P}[L_n(R;\phi) > e^{u}\Lambda_n(\phi, p_0)\ell_n(\phi, p_0)] \leq Ce^{-cu^2/\log(2+u)}, \tag{A.6}$$

for $u \geq u_0$. The same estimates hold for ψ, after changing p_0, c, C, u_0 by fixed amounts.

Proof. We check the dependence on ξ in the RSW and tail arguments of [DDDF20]. In the Gaussian comparison step of [DDDF20, Lemma 9 and Proposition 10], the independent centered Gaussian field added to the exponent has pointwise variance at most $v \leq C\xi^2$. Conditional on a crossing and its probability length measure μ, Jensen's inequality gives

$$\mathbb{E}\left[\left(\int e^{\Psi}\, d\mu\right)^a \middle| \mu\right] \leq e^{a^2 v/2}, \qquad a \geq 1.$$

For a logarithmic threshold $s \geq v$, take $a = s/v$ to obtain $e^{-s^2/(2v)}$. When a fixed comparison probability is chosen, v may therefore be replaced by the uniform upper bound $C\xi_*^2$. No lower bound on ξ is used. The finitely many conformal maps, rectangle shapes, subdivisions, and positive-association steps in the RSW argument are independent of ξ. Hence the RSW comparison probabilities and constants can be chosen uniformly for $\xi \in (0, \xi_*]$.

For the lower tail, the recursion in [DDDF20, Proposition 16] has the form

$$p_{j+1} = (Cp_j)^2, \qquad r_{j+1} = C^{-1} r_j \exp\{-C\xi\sqrt{\log(1/p_j)}\},$$

with the constants just fixed. Choose p_0 so small that $C^2 p_0 < 1$. Then

$$p_j \leq Ce^{-c2^j}, \qquad r_j \geq r_0 \exp\{-C(j+1) - C\xi 2^{j/2}\}.$$

Putting $2^{j/2}$ comparable to a real parameter gives a Gaussian lower tail at a threshold of the form

$$r_0 \exp\{-C - C\log(2+s) - C\xi s\}.$$

Since $\xi \leq \xi_*$, a fixed linear rescaling of s gives the first estimate of the lemma. The RSW comparisons and (A.3) transfer it from ψ to ϕ.

For the upper tail, use the proof of [DDDF20, Proposition 18]. The finite-range percolation part has a dependence range fixed by the cutoff for ψ. The only ξ-dependent terms are Gaussian comparison and coarse-field errors. By (A.1)–(A.2), for $1 \leq m \leq n$ the proof gives, with constants independent of $\xi \leq \xi_*$,

$$\mathbb{P}\left[\log \frac{L_n(R;\phi)}{\Lambda_n(\phi, p_0)\ell_n(\phi, p_0)} > C_0 2^{m/2}\sqrt{m+1}\right] \leq Ce^{-c2^m}. \tag{A.7}$$

Indeed, the passage from ψ to ϕ contributes at most $C\xi(\sqrt{m+1} + 2^{m/2})$ in the exponent, the coarse maximum contributes at most $C\xi m + C\xi 2^{m/2}\sqrt{m+1}$, and the comparison of fine and full lower

quantiles contributes at most $C\xi m + C\sqrt{m+1}$. Increasing C_0 once bounds all of these uniformly for $\xi \le \xi_*$.

Put $u_m = C_0 2^{m/2}\sqrt{m+1}$. For $u \le u_n$, choose the largest m with $u_m \le u$. Above a fixed starting value,

$$2^m \ge c\frac{u^2}{\log(2+u)}.$$

For $u > u_n$, one has $n \le C\log(2+u)$. Also

$$\ell_n(\phi, p_0) \ge e^{-C\xi(n+\sqrt{n+1})}$$

by comparison with the infimum of the field. Along a fixed straight crossing, for $a \ge 1$,

$$\mathbb{E}[L_n(R;\phi)^a] \le C_R^a \exp\{a^2\xi^2 n \log 2/2\}.$$

Markov's inequality, with a a sufficiently small fixed multiple of $u/(\xi_*^2(n+1))$, gives an upper bound $Ce^{-cu^2/(\xi_*^2(n+1))}$ after increasing the fixed starting value of u. Since $n \le C\log(2+u)$ in this case, this has the form in (A.6). The finitely many remaining small-scale cases follow directly from (A.1). This proves the lemma. □

A.3 A uniform bound for the quantile ratio

Let π_n^k be the set of dyadic k-squares visited by an internal unit-square crossing geodesic for $\psi_{0,n}$. It contains at least $c2^k$ distinct squares. If $w_P = e^{\xi\psi_{0,k}(P)}$, with the field evaluated at the center of P, then

$$R_{n,k} := \frac{\sum_{P\in\pi_n^k} w_P^2}{\left(\sum_{P\in\pi_n^k} w_P\right)^2} \le C2^{-k}e^{2\xi\sup_V|\psi_{0,k}|}. \tag{A.8}$$

Equations (A.1)–(A.2) imply

$$\|R_{n,k}\|_2 \le C\exp\{-k\log 2 + C\xi k + C\xi^2 k\}.$$

Choose $\xi_* > 0$ smaller, if necessary, so that

$$\|R_{n,k}\|_2 \le Ce^{-c_0 k}, \qquad n \ge k, \tag{A.9}$$

for deterministic $c_0 > 0$ independent of n and $\xi \le \xi_*$. This is the form of Condition (T) from [DDDF20] which is used in the Efron–Stein argument.

Lemma A.2. *After decreasing ξ_* if necessary, there is $C < \infty$ such that*

$$\sup_{\xi\in(0,\xi_*]}\sup_{n\ge 0}\Lambda_n(\phi,p_0) + \sup_{\xi\in(0,\xi_*]}\sup_{n\ge 0}\Lambda_n(\psi,p_0) \le C.$$

Consequently, for every fixed rectangle shape and marked direction,

$$\mathbb{P}[L_n(R;\phi)/\lambda_n < e^{-u}] \le Ce^{-cu^2}, \qquad \mathbb{P}[L_n(R;\phi)/\lambda_n > e^u] \le Ce^{-cu^2/\log(2+u)}, \quad u \ge 2, \tag{A.10}$$

and the same is true with ϕ replaced by ψ. In particular, every fixed positive and negative crossing moment is bounded uniformly in n and $\xi \le \xi_*$.

Proof. Fix the quantile probability p_0 in Lemma A.1 sufficiently small that the estimates for ϕ can be used at probability $2p_0$. Write

$$V_n = \operatorname{Var} \log L_n([0,1]^2;\psi), \qquad \Lambda_n = \Lambda_n(\psi, p_0).$$

We now spell out the constants in the Efron–Stein recurrence from the proof of [DDDF20, Theorem 20]. Resampling the coarse field through scale k contributes at most

$$C\xi^2 k \le C\xi_*^2 k$$

to V_n. For the fine contribution, the deterministic detour and crossing inequalities in [DDDF20, equations (5.61)–(5.67)] bound it by

$$Ck^a \mathbb{E}[R_{n,k} B_{n,k}^2 G_k]$$

for a deterministic exponent a. Here $B_{n,k}$ is the ratio of the largest relevant fine-field long-rectangle crossing to the smallest relevant fine-field short-rectangle crossing, and G_k is the exponential of the coarse oscillation and the comparison-field errors.

There are at most $C4^k$ fine rectangles. A union bound, integration of the two tails in Lemma A.1, and (A.4) give, for any fixed $\nu \in (1/2, 1)$,

$$\|B_{n,k}\|_8 \le C\Lambda_{n-k} e^{C(k+1)^\nu}. \tag{A.11}$$

Fix the finite-range cutoff exponent $\varepsilon_0 \in (0, 1/2)$. The enlarged blocks have diameter at most $Ck^{\varepsilon_0}2^{-k}$. The scaled gradient bound following (A.1) and the comparison estimate (A.2) give

$$\|G_k\|_4 \le C \exp\{C\xi k^{1/2+\varepsilon_0} + C\xi^2 k^{2\varepsilon_0} + C\xi + C\xi^2\} \le C e^{C(k+1)^{\nu'}},$$

where $\nu' < 1$ is fixed and larger than both ν and $1/2 + \varepsilon_0$. No independence of $R_{n,k}$, $B_{n,k}$, and G_k is used. Hölder's inequality and (A.9) give

$$\mathbb{E}[R_{n,k} B_{n,k}^2 G_k] \le \|R_{n,k}\|_2 \|B_{n,k}\|_8^2 \|G_k\|_4 \le C e^{-c_0 k + Ck^\nu + Ck^{\nu'}} \Lambda_{n-k}^2.$$

The fixed power k^a and the sublinear exponential can be absorbed into e^{-ck} after increasing a deterministic k_1. Thus there are fixed $A, c, k_1 > 0$ such that

$$V_n \le Ak + e^{-ck}\Lambda_{n-k}^2, \qquad n \ge k \ge k_1, \tag{A.12}$$

uniformly for $\xi \le \xi_*$. This is exactly the recurrence needed below; in particular, no constant in it depends on ξ.

For a random variable Z of finite variance, the distance between its lower p_0- and $(1-p_0)$-quantiles is at most a constant, depending only on p_0, times $\sqrt{\operatorname{Var} Z}$. Applying this to $Z = \log L_n([0,1]^2;\psi)$ gives

$$\frac{\overline{\ell}_n(\psi, p_0)}{\ell_n(\psi, p_0)} \le e^{a\sqrt{V_n}}. \tag{A.13}$$

The finite-scale variance estimate in the proof of [DDDF20, Lemma 23] gives $V_j \le C(j+1)$ uniformly for $\xi \le \xi_*$; here the only parameter-dependent contribution is bounded by $C\xi^2(j+1)$. Hence

$$\Lambda_k \le e^{B\sqrt{k+1}}$$

with fixed B.

Choose one large deterministic integer $k \geq k_1$ and set

$$R_0 = \max\{e^{B\sqrt{k+1}}, e^{a\sqrt{2Ak}}\}.$$

Since $\log R_0 = O(\sqrt{k})$, increasing k if necessary gives $e^{-ck}R_0^2 \leq Ak$. The bound $\Lambda_j \leq R_0$ holds for $j \leq k$. If it holds through index $n-k$, the recurrence gives $V_n \leq 2Ak$, and (A.13) gives the same bound R_0 for the new quantile ratio. Induction gives $\Lambda_n \leq R_0$ for every n.

The comparison (A.4) gives the same conclusion for ϕ. Finally, Lemma A.1 and (A.4) show that every fixed quantile is uniformly comparable to the median λ_n, which gives (A.10). Integrating the tails gives the final moment statement. □

A.4 Crossing medians at different scales

The next statement is the uniform version of the weak multiplicativity estimate from [DDDF20, Proposition 26].

Lemma A.3. *There is a constant C, independent of $\xi \in (0, \xi_*]$, such that*

$$e^{-C\sqrt{k}}\lambda_n\lambda_k \leq \lambda_{n+k} \leq e^{C\sqrt{k}}\lambda_n\lambda_k, \qquad n, k \geq 1. \tag{A.14}$$

Proof. We indicate the two bounds separately in order to make the uniformity clear.

For the upper bound, use the coarse crossing at scale k and replace each portion crossing a coarse block by the corresponding collection of fine long-rectangle crossings, as in the proof of [DDDF20, Proposition 26]. By Lemma A.2, the first and second moments of the required normalized fine crossings are bounded by fixed constants, uniformly in n, k, and $\xi \leq \xi_*$. The oscillation of the coarse field on a k-block is controlled by the scaled gradient estimate following (A.1). A union bound over the relevant blocks shows that, outside an event of fixed small probability, the resulting multiplicative error is at most $e^{C\sqrt{k}}$. Taking expectation for the fine replacements and then applying Markov's inequality at a fixed level gives

$$\lambda_{n+k} \leq e^{C\sqrt{k}}\lambda_n\lambda_k.$$

All constants used here come from (A.1), (A.10), and the fixed finite collection of rectangle shapes.

For the lower bound, every coarse block visited by a fine crossing contains a crossing of one of a fixed collection of short rectangles. There are at most $C4^k$ possible translated rectangles at coarse scale. The Gaussian lower tail in (A.10), with threshold of order $\sqrt{k}$, shows that with fixed probability arbitrarily close to one every such fine crossing is at least $e^{-C\sqrt{k}}\lambda_n$. The coarse oscillation contributes another factor of this form. Separately, take a fixed lower quantile of the coarse crossing; by Lemma A.2, it is bounded below by a fixed multiple of λ_k. Choose the fixed probabilities so that the intersection of the coarse-crossing event and the two error events has probability greater than $1/2$. On this intersection,

$$L_{n+k}([0,1]^2; \phi) \geq e^{-C\sqrt{k}}\lambda_n\lambda_k.$$

This yields the lower median bound. The constants are again independent of $\xi \leq \xi_*$. □

Set $f(k) = \log \lambda_k$. Taking $n = k$ in (A.14) gives

$$|f(2k) - 2f(k)| \leq C\sqrt{k}.$$

Divide by $2k$ and telescope along $2^j k$. The resulting series is bounded by $C/\sqrt{k}$. For each fixed $\xi = \xi_\gamma$, the LFPP exponent identifies the limiting slope as $-(1-\xi_\gamma Q_\gamma)\log 2$; see [DG20, Theorem 1.5] and [DDDF20, Proposition 26]. The error estimate from Lemma A.3 is already uniform, so no uniform rate in this fixed-parameter identification is needed. Hence

$$|\log \lambda_k + k(1-\xi_\gamma Q_\gamma)\log 2| \le C\sqrt{k}, \qquad e^{-C\sqrt{k}} 2^{k(1-\xi_\gamma Q_\gamma)} \le \frac{\lambda_{n-k}}{\lambda_n} \le e^{C\sqrt{k}} 2^{k(1-\xi_\gamma Q_\gamma)}, \quad 1 \le k \le n. \tag{A.15}$$

For the ratio bounds, apply (A.14) with indices $n-k$ and k and then use the first estimate. The case $n = k$ uses $\lambda_0 = 1$.

A.5 Upper Hölder moments and inverse distance moments

Fix $p \ge 1$ and $\nu \in (1/2, 1)$. At scale 2^{-k}, let $\mathcal{R}_k$ be a family of at most $C4^k$ fixed-shape rectangles in a bounded region. Scaling, a union bound in (A.10), and integration of the upper tail give

$$\left\| \max_{R \in \mathcal{R}_k} L(R; \phi_{k,n}) \right\|_p \le C_p 2^{-k} \lambda_{n-k} e^{C_p (k+1)^\nu}, \qquad 0 \le k \le n. \tag{A.16}$$

The first index k means that only white-noise scales between k and n are included. The formula includes $k = n$, when the fine field is zero. The fine field is independent of $\phi_{0,k}$. Multiply by $e^{\xi \sup_V \phi_{0,k}}$, use (A.1) and (A.15), and set $\xi = \xi_\gamma$. Then

$$\left\| \lambda_n^{-1} \max_{R \in \mathcal{R}_k} L(R; \phi_{0,n}) \right\|_p \le C_p 2^{-k\{\xi_\gamma Q_\gamma - C_0(\xi_\gamma + p\xi_\gamma^2)\}} e^{C_p (k+1)^\nu}. \tag{A.17}$$

For fixed p, choose γ_p so small that the expression in braces is at least $7/8$. Absorb the sublinear exponential into a constant times $2^{k/8}$ to obtain $C_p 2^{-3k/4}$.

The rectangle-chaining construction in [DDDF20, Propositions 27–28] connects pairs at Euclidean separation at most 2^{-k} using a fixed number of these crossings at each scale $i \ge k$, together with straight terminal pieces of length $O(2^{-n})$. The paths remain in a fixed slightly larger rectangle V. Minkowski's inequality gives

$$\left\| \lambda_n^{-1} \sup_{\substack{x,y \in K \\ |x-y| \le 2^{-k}}} D_{\phi,n}(x, y; V) \right\|_p \le C_{K,p} 2^{-3k/4}, \qquad 0 \le k \le n, \tag{A.18}$$

for a fixed rectangle K strictly inside V. The terminal term is bounded by $C2^{-n} e^{\xi \sup_V \phi_{0,n}}$, which satisfies the same estimate by (A.1) and (A.15). Below the cutoff, straight segments bound the $1/2$-Hölder quotient by

$$C\lambda_n^{-1} e^{\xi \sup_V \phi_{0,n}} 2^{-n/2}.$$

Its L^p norm is uniformly bounded by the same estimates. Multiply (A.18) by $2^{k/2}$ and sum in k to obtain the upper Hölder moment. Finite covers by translates of fixed-size rectangles handle arbitrary compact sets. For endpoint pairs not lying in one rectangle, use finite chains through a connected cover of a containing box.

For lower bounds, cover K by squares of radius $r \asymp \delta$, chosen so small that a point at Euclidean distance at least δ from a point in an inner square lies outside its concentric square of radius $4r$. Every path between such points contains an internal crossing of one of the rectangles

$$[r, 4r] \times [-4r, 4r],$$

or a rotation and translation of such a rectangle. Use the first exit through an outer face and the last preceding hit of its inner parallel line. Between these times the path remains in the corresponding rectangle. This also applies to paths which later leave the containing compact region.

Write $r \asymp 2^{-k}$. If $n \geq k$, a crossing is at least its fine crossing times $e^{\xi \inf_V \phi_{0,k}}$. Use the lower ratio bound in (A.15), namely

$$\frac{\lambda_n}{\lambda_{n-k}} \leq 2^{-k(1-\xi_\gamma Q_\gamma)} e^{C\sqrt{k}}.$$

Together with (A.1) and (A.10), this gives

$$\|\lambda_n / L_n(R;\phi)\|_p \leq C_p e^{C_p(k+1)}. \tag{A.19}$$

There are at most $C_K(1+\delta^{-2})$ rectangles. Taking the maximum inverse length gives a bound $C_{K,p}\delta^{-c_p}$ for the inverse of λ_n^{-1} times the global separated distance. If $n < k$, use the Euclidean crossing length times $e^{\xi \inf \phi_{0,n}}$ directly; (A.1) and $n < k$ give the same polynomial bound. No convergence of internal metrics is used in this step.

A.6 Comparison with the pinned GFF

Let σ be uniform probability measure on $\partial B_1(0)$ and $b_t = \sigma * p_{t/2}$. On a white-noise probability space, the distribution

$$h(z) = \sqrt{\pi} \int_0^\infty \int_{\mathbb{R}^2} [p_{t/2}(z-w) - b_t(w)]\, W(\,dw,\, dt) \tag{A.20}$$

has covariance (1.1). This follows by subtracting the unit-circle average in both variables from the heat-kernel representation of the logarithmic kernel. The integral is interpreted distributionally near $t = 0$.

Use the notation of [GM21], $h_\varepsilon^* = p_{\varepsilon^2/2} * h$. Change variables to $u = t + \varepsilon^2$ in (A.20), and let $\widetilde{W}$ be the shifted white noise. Couple h_ε^* to

$$\phi_\varepsilon(z) = \sqrt{\pi} \int_{\varepsilon^2}^1 \int_{\mathbb{R}^2} p_{u/2}(z-w)\, \widetilde{W}(\,dw,\, du),$$

which has the required DDDF law. Their difference is $C_\varepsilon + V_\varepsilon(z)$, where

$$C_\varepsilon = -\sqrt{\pi} \int_{\varepsilon^2}^1 \int_{\mathbb{R}^2} b_{u-\varepsilon^2}(w)\, \widetilde{W}(\,dw,\, du),$$

$$V_\varepsilon(z) = \sqrt{\pi} \int_1^\infty \int_{\mathbb{R}^2} [p_{u/2}(z-w) - b_{u-\varepsilon^2}(w)]\, \widetilde{W}(\,dw,\, du).$$

The bound

$$\|b_t\|_2^2 = \iint p_t(v-v')\sigma(\,dv)\sigma(\,dv') \leq Ct^{-1/2}, \qquad 0 < t \leq 1,$$

shows that C_ε has uniformly bounded variance. For z in a fixed compact set and $u \geq 1$, when $\varepsilon^2 \leq 1/2$,

$$\|p_{u/2}(z-\cdot) - b_{u-\varepsilon^2}\|_2 \leq C_K u^{-1}, \qquad \|\partial_z^\ell p_{u/2}(z-\cdot)\|_2 \leq C_\ell u^{-(|\ell|+1)/2}, \quad |\ell| \geq 1.$$

Here ℓ is a multi-index in the two real coordinates. These estimates give uniform variance bounds for V_ε and all of its spatial derivatives on compact sets. Sobolev embedding bounds the expected supremum, and Gaussian concentration gives

$$\mathbb{P}[\|h_\varepsilon^* - \phi_\varepsilon\|_{L^\infty(K)} > t] \leq C_K e^{-c_K t^2}. \tag{A.21}$$

A separate coupling for each cutoff suffices; the error need not be independent of ϕ_ε.

A.7 Passage to the continuum metric

Define the whole-plane LFPP metric by

$$D_h^{\varepsilon,\gamma}(z,w) := \inf_{P:z\to w} \int_0^1 e^{\xi_\gamma h_\varepsilon^*(P(t))} |P'(t)|\, dt,$$

where paths are piecewise continuously differentiable and lie in $\mathbb{C}$. Let

$$a_\varepsilon^\gamma := \operatorname{med}\left(\inf_{s,t\in[0,1]} D_h^{\varepsilon,\gamma}(is, 1+it) \right). \tag{A.22}$$

For $\varepsilon = 2^{-n}$, the comparison with the white-noise field gives the estimates above for $\lambda_n^{-1} D_h^{2^{-n},\gamma}$. The comparison (A.21), Hölder's inequality at larger fixed moment orders, and Section A.5 transfer the upper Hölder and lower separated-distance moment bounds to $\lambda_n^{-1} D_h^{2^{-n},\gamma}$. For the upper bound, use internal paths in fixed rectangles. For the lower bound, use only the compact rectangle subpaths described above. No comparison of fields at infinity is needed.

Write $C_{\varepsilon,\gamma}$ for the crossing variable inside (A.22). Then

$$\inf_{\substack{z,w\in[0,1]^2 \\ |z-w|\geq 1}} D_h^{\varepsilon,\gamma}(z,w) \leq C_{\varepsilon,\gamma} \leq D_h^{\varepsilon,\gamma}(0,1).$$

Thus $\lambda_n^{-1} C_{2^{-n},\gamma}$ has uniformly bounded positive and negative moments. Markov's inequality at one fixed moment order gives

$$c\lambda_n \leq a_{2^{-n}}^\gamma \leq C\lambda_n \tag{A.23}$$

uniformly in n and sufficiently small γ. All established moment estimates therefore remain true with $(a_{2^{-n}}^\gamma)^{-1}$ in place of λ_n^{-1}.

Now fix γ. By [GM21, Theorem 1.1], $(a_\varepsilon^\gamma)^{-1} D_h^{\varepsilon,\gamma}$ converges locally uniformly in probability to D_h^γ as $\varepsilon \to 0$. Along an a.s. convergent subsequence, the upper Hölder seminorm is lower semicontinuous, and the infimum over the compact separated-pair set is continuous. Fatou's lemma transfers the bounds above to D_h^γ. The cutoff is sent to zero with γ fixed, after all estimates have been made uniform for sufficiently small γ. This proves Lemma 2.1.

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